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Early Stage Warning and Monitoring System for Floods and Landslides in Rwanda

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Abstract

Climate hazards such as floods and landslides are increasing across the world in both frequency and severity. These events result in significant human, environmental, and economic losses each year. Despite advances in environmental monitoring, the lack of integration between space-derived data and local ground-based measurements remains a key limitation in effective early warning systems. A solution to this amalgamates a sensor network monitoring and early warning system with Earth Observation (EO) data and low-cost ground-based sensors measuring soil moisture, ground movement and temperature. Such instruments can be incorporated into solar powered devices to collect and transmit data helping to build a network of sensors and can contribute to improved climate models, while also warning communities about localised developing hazardous conditions. Integrating *in-situ* ground measurements with EO data could enable key variables, including the influence of human activity on regional vulnerability, to be more accurately analysed. This project supports the development and refinement of early warning systems and machine learning models, to improve predictive accuracy for hazard events. The results aid the identification of circumstances which cause EO data processing times to limit rapid responses, and illustrates where ground-based sensors can provide faster localised information, particularly to more rural communities, strengthening resilience to climate hazards. Two ground hardware configurations are explored; firstly, a modular system constructed from off-the-shelf Arduino components for educational and community-based deployment is demonstrated, and secondly, a compact integrated design which can be better suited to distribution at scale (for example as part of governmental EO programmes). Rwanda provides a representative case study of this challenge. Its steep terrain, high rainfall variability and dispersed rural communities create conditions in which landslides and flash floods can occur with limited early warning. The work presented builds on the research first conducted through the 2025 edition of the International Space University Space Studies Program examining environmental risk, economic vulnerability, and the role of space-based technologies in disaster management.

Keywords: Earth Observation, Ground Sensors, Early Warning System, Machine Learning, Satellite Data

1. Introduction

Floods and landslides pose a significant threat to communities worldwide, resulting in loss of life, damage to infrastructure, disruption of livelihoods, and wider economic impacts. The World Health Organization (WHO) estimates that between 1998 and 2017, 4.8 million people were affected by landslides, while floods impacted over 2 billion lives globally in the same time frame [1,2]. A key component in mitigating the damages caused by these natural catastrophes is the ability to identify areas at risk and detect hazardous conditions before and as they develop. This challenge is particularly relevant in Rwanda, where rural communities struggle to cope with recently shifted rainfall patterns and the increased frequency and intensity of extreme weather events due to climate change, ultimately

leading to floods and landslides [3].

Earth Observation (EO) satellites provide an increasingly powerful means of addressing this problem [4]. Optical, radar, and elevation data can characterise terrain, vegetation, land use, and surface change across vast areas that would be impractical to monitor comprehensively from the ground, providing important inputs to flood and landslide susceptibility models. However, revisit intervals, cloud-related constraints, spatial resolution, and data-processing times can limit the availability of sufficiently local and timely observations during rapidly developing events [5,6].

In-situ data collection using ground sensors provides a complementary source of information. Measurements such as soil moisture content, temperature, humidity and ground movement can be collected locally at substantially

higher temporal frequency than periodic satellite observations, providing direct observations of changing conditions [7]. However, ground sensor networks suffer from large-scale implementation and maintenance issues. Combining these measurements with the wide-area coverage of EO therefore provides an opportunity to overcome limitations associated with the use of either approach independently.

For these measurements to contribute effectively to disaster-risk reduction, they must ultimately form part of an Early Warning System (EWS) capable of converting observations into timely and actionable information. This requires the integration of sensor measurements, predictive modelling, and communication infrastructure with the communities exposed to the hazard. In regions where extensive monitoring infrastructure may be economically or logistically impractical, low-cost, solar-powered sensors and long-range wireless communication offer a potential means of extending monitoring to otherwise underserved locations.

This work investigates a hybrid approach to flood and landslide monitoring, using Rwanda as a case study. Satellite EO is considered across the disaster cycle, with particular attention to temporal limitations and anthropogenic drivers of hazard risk. An existing machine-learning flood-susceptibility model for the Nyabarongo catchment – initially proposed by Nzabonantuma *et al.* [8] – is extended to investigate built-up land, vegetation change, and wetness decline as additional predictors. In addition, a complementary low-cost ground-sensing platform is developed to measure soil moisture content, atmospheric conditions, and ground movement, with modular and integrated hardware configurations considered for community-based and larger-scale deployment, respectively. Building on these components, this paper proposes an integrated EWS that combines the regional coverage of EO, local and temporally dense ground measurements, and predictive modelling. The proposed architecture classifies four hazard levels and associates each with recommended response actions, providing an affordable and scalable solution for communities exposed to floods and landslide hazards.

2. Current Satellite Technologies for Earth Observation of Floods and Landslides

Satellite EO provides a continuous, wide-area view of land surface that is highly challenging to achieve through ground-based monitoring alone. A combination of optical, radar, elevation, and light-based data is relevant to flood and landslide risk. Optical multispectral sensors, such as the Landsat series and Sentinel-2 capture land cover, veg-

etation health and land-use change that help characterise a catchment's susceptibility to floods and slope instability [9]. Synthetic Aperture Radar (SAR) sensors, such as Sentinel-1, can penetrate cloud cover and darkness, making it particularly valuable for mapping flood extent during and immediately after a storm when optical imagery is often unusable. SAR-derived techniques including Interferometric Synthetic Aperture Radar (InSAR) and Differential SAR Interferometry (DInSAR), can also detect small ground displacements associated with slope instability and landslide precursors [9]. Elevation data from sources such as the Shuttle Radar Topography Mission (SRTM) digital elevation model underpins terrain-based factors such as slope, aspect, and topographic wetness that are common inputs to flood and landslide susceptibility models [10]. Together, these satellite types span the full disaster cycle, offering the technical foundation for before, during and after event monitoring methods discussed in this section.

2.1 Pre-Event Susceptibility Modelling

Section 3 carries out an in-depth review and analysis of the current literature and techniques of machine learning models used for pre-event susceptibility modelling.

2.2 During-Event - Real-Time Detection

Real-time flood monitoring is largely absent from the published literature on Rwanda. Sentinel-1 SAR has been used in the country for agricultural monitoring, but minimal published work applies it to flood detection, and an operational “during-flood” monitoring system does not exist at the time of writing. This is despite the existing methods being proven elsewhere. Sentinel-1 SAR is used globally to map flood extent through cloud cover and Global Precipitation Measurement Integrated Multi-satellite Retrievals (GPM IMERG) to provide near-real-time rainfall data combined with a rainfall trigger and flood-extent monitoring [11, 12]. Google Earth Engine (GEE), a cloud-based platform that provides access to large collections of satellite and EO data for processing, analysis and environmental monitoring then provides the cloud-processing capacity needed for rapid analysis [13]. A comparable global system, Global Forest Watch, already combines Landsat-based classification with near-real-time deforestation alerts for Peru, the Democratic Republic of Congo, and Borneo, illustrating the kind of standing operational infrastructure that flood monitoring in Rwanda currently lacks [14]. Satellite-based flood forecasting has been used in Rwanda with a 72% hit rate and a three-hour lead time, though this has not been developed into a standing op-

erational system [15]. A useful comparison for what an operational near-real-time system in Rwanda could look like is Brazil's DETER-R deforestation warning system, which layers cloud-penetrating Sentinel-1 SAR data onto an existing optical-image system and uses forest masks with optical validation [12]. However, even this mature system carries a day-long delay between disturbance and an issued warning. Attribution of the type of disturbance such as smallholder agriculture, road development, selective logging, or mining has also been demonstrated using a Convolutional Neural Network (CNN) classifier in the Democratic Republic of Congo, but accuracy there improves with the length of the post-disturbance observation window, between four and six months, illustrating a direct trade-off between speed and classification accuracy [11].

2.3 Post-Event - Recovery and Resilience

Post-event monitoring can be divided into three phases with different data needs. Firstly, the immediate phase covering the hours to days following an event can be difficult to observe due to persistent cloud cover. However, SAR imagery can provide observations through cloud cover making it particularly valuable during this phase, although observations are constrained by satellite revisit times of approximately five days [16]. Secondly, a short-term phase coming in the days after the event can draw on high-resolution drone imagery. Finally a long-term phase, months after the event can use night-time lights (NTL) together with Digital Elevation Models (DEM) to understand future flood risk. While NTL data exists for Rwanda and has been used to assess electrification, it is biased towards urban areas. Applying NTL to monitor flood recovery may be effective in Kigali, however the risk and impact of floods and landslides is often greater in remote areas where NTL is less effective or unavailable [17]. The NTL visibility gap for rural flood impacts results in the technique being less applicable. A United States-based study validated flood simulations by comparing drone-derived DEMs collected before and after a flood event, identifying areas of higher water depth to target for restoration [18]. This raises an open question of whether satellite-derived DEM change detection could offer a similar capability in regions without routine access to drone survey capacity. It is evident that additional work is needed to ensure post-flood recovery is equitable across areas of differing development levels, since better-developed areas tend to recover faster and could otherwise draw a disproportionate share of monitoring attention and resources [19].

2.4 Human Drivers of Flood and Landslide Risk

Floods and landslides are increasingly driven and amplified by human modification of the land surface [20]. EO offers a route to monitoring underlying human drivers and the resulting flood and landslide events. This subsection reviews the human activities that most directly contribute to flood and landslide risk. Deforestation, urbanisation, and mining each alter how water infiltrates, drains, and accumulates across a catchment, changing the frequency and severity of downstream hazards.

2.4.1 Deforestation

Deforestation degrades and loosens soil, and this loose soil can move down slopes and trigger landslides. The same erosion reduces channel water depth, which in turn increases flood risk; other downstream effects include the loss of nutrient-rich topsoil, reduced agricultural yields and disruption of the water cycle leading to drought. In Rwanda, an analysis of hydrological change between 1991 and 2020 found that forestland fell from 49.9 to 11.8% of the study area while cropland rose from 33.2 to 72.0%, with urban expansion and agricultural intensification identified as the principal drivers [20]. This loss of forest cover is reflected in reduced satellite-derived vegetation indices such as Normalised Difference Vegetation Index (NDVI), which correspond to lower soil infiltration capacity and higher surface run-off, directly increasing flood risk in deforested catchments [8]. Forest conservation in Rwanda is complicated by the fact that fuelwood provides 90% of the country's energy, so any conservation policy is debated against competing priorities of energy transition, poverty alleviation, food security, and regional economic development [21]. Conflict-driven redistribution of people has also been shown to increase pressure on forest resources in western Rwanda, though distinguishing plantations from natural forest in satellite imagery remains difficult and can skew deforestation estimates [21].

2.4.2 Urbanisation

Urban expansion converts permeable land into impervious surface, directly reducing infiltration and increasing run-off. In Kigali, built-up area grew by 226.5% from 1987 to 2019 with impervious surface expanding at an average annual rate of 3.7%, accelerating to 4.5% per year after 2004 [22]. The causal relationship between this kind of change and flood magnitude has been shown as an average 3.3% increase in annual flood magnitude per 1% increase in impervious basin cover [23]. Urbanisation in Kigali has also come at the expense of natural flood buffers.

Wetland area experienced a 28% loss in six years from 2013, driven by urban encroachment, agricultural conversion, and drainage for development [24]. In response, the Second Rwanda Urban Development Project (RUDP II) is restoring roughly 500 hectares of urban wetland, intended to benefit 220,500 people in flood-prone neighbourhoods [24]. Despite this evidence, current Rwandan flood susceptibility models do not include impervious surface as a direct predictor [8]. Land cover is represented only indirectly through NDVI or a Curve Number derived from land-use/soil look-up tables, and the Curve Number was found to rank last of ten predictive factors in a Shapley Additive Explanations (SHAP) feature-importance analysis [8,25]. This is likely a consequence of using a proxy that collates fine-grained urban patterns into a single run-off coefficient rather than measuring impervious cover directly.

2.4.3 Mining

Mining activity alters land cover and can destabilise ground. A combination of vegetation and wetland loss mirrors the deforestation pathway to increased flood risk, while the associated ground instability around excavation and waste-disposal sites can independently trigger landslides, particularly where tailings-dams or steep excavation slopes are involved [16,26,27]. Beyond land-cover change, mining also induces ground instability [16]. InSAR is established as a companion to in-situ instruments such as ground-based radar and geotechnical sensors for monitoring subsidence and tailings-dam instability, and long-term sequences of SAR data processed through multi-temporal DInSAR provide accurate measurement of the resulting ground displacement [26].

2.5 Data Latency and Temporal Constraints

A recurring theme across all three monitoring phases is that satellite revisit frequency, cloud cover and processing pipelines currently limit how quickly human drivers and flood events can be detected and acted upon. Table 1 summarises the resolution, revisit cadence, and reported delays or constraints identified across the reviewed literature.

These figures show that while individual sensors can revisit an area every five to sixteen days, end-to-end operational delay is often much longer once processing, cloud contamination, and validation steps are included, as illustrated by the 18 - 24 day warning delay in Brazil's DETER system, despite it already combining SAR and optical data [12]. This is demonstrated further by Rwanda's current satellite based flood forecasting system with a 72% hit rate with a 3-hour lead time proving near-real-time fore-

casting is achievable, although not yet optimised [15]. Atmospheric effects can also remove usable data at precisely the time of greatest risk, as seen in Ghana where roughly a quarter of Sentinel-1 acquisitions during the rainy season were unusable [29]. This indicates that closing the gap between proven near-real-time capability and a standing operational EWS is a challenge for both data-pipeline, validation, and sensing. The growing availability of satellite imagery combined with mature classification algorithms and cloud-based processing now allows analysis to run at global scale [30]. Furthermore, low-cost ground sensors can complement Earth observation by providing continuous local measurements, helping to overcome their inherent temporal limitations. There is an opportunity to improve flood and landslide EWSs by integrating these complementary technologies into a single operational system.

3. Machine Learning Model Integration

This section first reviews existing machine learning approaches to flood and landslide susceptibility modelling in Rwanda before presenting an extension of an existing baseline model to incorporate the human-driver variables identified in Section 2.4, together with the outcomes of the conducted experiments.

In Rwanda's Nyabarongo catchment, dense population growth and rapid urbanising of steep terrain have worsened the already significant flood and landslide hazard [8,22]. Predicting where and when these events are likely to occur is central to reducing their impact. Machine learning offers a means of combining the terrain, land cover and hydrological variables discussed in Section 2 into a single predictive framework. This translates EO data into a spatially explicit measure of susceptibility that can inform both long-term planning and the EWS, to be discussed in Section 5.

A range of existing machine learning techniques have been applied to flood susceptibility prediction. Reported results include a Gradient Boosting model that predicted 90% of flood events reported between 2013 and 2022, and a CNN combined with a Deep ResNet architecture achieving close to 95% accuracy [31,32]. Predictive factors typically include elevation, slope, aspect, profile curvature, distance from rivers and roads, NDVI, the Normalised Difference Soil Index (NDSI), the Topographic Wetness Index (TWI), and rainfall. Two Rwanda-specific studies illustrate this approach directly. A 2019 logistic regression model achieved an area under the curve (AUC) of 0.798 using 153 historical flood points (2012-2017), an SRTM-derived 30m DEM, Landsat 8-derived NDVI and NDSI, and CHIRPS rainfall [25]. A 2025 study applied various machine learning models such as Random Forest, SVM, and XGBoost

Table 1: Satellite data sources and temporal considerations

Data system	Resolution	Revisit / cadence	Significant considerations
Landsat (5/7/8/9)	30 m	16 days	Continuous archive since 1984
Sentinel-2 MSI	10–20m	5 days	Cloud cover constrains usable acquisitions
Sentinel-1 SAR	5–20m	~5 days	Penetrates cloud but atmospheric issues [28]
DETER-R (Brazil) NRT deforestation warning	—	18 – 24 days	SAR + optical fusion; delay between disturbance and issued warning [12]
InSAR ground-deformation monitoring (mining)	Up to 1 × 1 m (commercial, small area)	Several days (large-area sensors)	Acquisition frequency limit effectiveness for Early Warning Systems [16, 26]

to 151 flood points and achieved an AUC of up to 0.968, with distance to river, TWI, and rainfall ranking as the most influential predictors [8]. However, neither model included impervious surface percentage (ISP) directly, even though SHAP-based studies of urban catchments have repeatedly ranked impervious surfaces among the top three predictors of urban flood risk [33, 34], indicating a specific, addressable gap in Rwanda’s pre-event models.

3.1 Extending the Baseline Model

The baseline model reproduces the ten-variable Random Forest implementation of Nzabonantuma *et al.* [8], built directly on the authors’ publicly available code, with additional variables incorporated. The baseline’s statistic, categorical land use inputs do not directly capture the physical properties and temporal dynamics of human land modification discussed in Section 2.4. Three additional variables were investigated to address this limitation, as follows:

First, ISP was investigated alongside the Curve Number (CN) in urban and peri-urban catchments. As discussed in Section 2.4, impervious surface expansion has been shown to increase flood magnitude [23], and Kigali’s impervious surface area and associated surface run-off potential have grown substantially in recent decades [22], motivating the inclusion of a direct impervious surface predictor in the model. A suitable 2023 ISP product could not be sourced directly without introducing an additional classification model; therefore, the extended model developed in this project used the Built Percentage (BuiltPct), the proportion of each cell classified as built-up land, as the closest available proxy.

Second, a dynamic NDVI change variable was investigated in place of static NDVI to capture the rate of vegetation loss and land-use conversion that precedes many

landslide events, rather than vegetation greenness at a single point in time. Human-driven land modification is a well-documented contributor to landslide risk, particularly in the case of Rwanda. Based on an inventory of over 4,600 mapped landslides across three major rainfall-triggered events in north-west Rwanda, Sibomana *et al.* [35] found that landslides were approximately three times more likely to occur on terraced human-modified hill-slopes than on non-terraced ones.

Third, a Wetness Decline (WetDeclPct) variable was investigated to capture drying of previously wet land in and around Kigali. As discussed in Section 2.4, wetland loss in this area has directly reduced natural flood buffering capacity [24]. However, the operational variable identifies a continuous spectral drying signal rather than confirmed ecological wetland destruction and is named accordingly rather than “wetland loss”.

3.2 Data and Methods

The following describes the data preparation and satellite-processing workflow developed in this project to derive BuiltPct, NDVI change, and WetDeclPct; it is independent of Nzabonantuma *et al.*’s original baseline methodology. The Nyabarongo study area was represented as a grid of 213,908 polygon cells, each approximately 200 m × 200 m, along with the original set of 302 flood and non-flood sample points used to train the baseline model [8]. The coordinate reference system of both layers was checked for consistency before processing. Because processing satellite imagery against the full set of individual grid polygons would have been unnecessarily complex, a single dissolved study area boundary was created in QGIS [36] using the dissolve tool and used to spatially restrict all subsequent satellite processing to the Nyabarongo catchment.

GEE was used to process satellite imagery, selected for its free access to large EO archives, direct access to Sentinel-2 imagery, cloud-based processing, and ability to process multi-year imagery without downloading individual satellite scenes. The dissolved study area boundary was uploaded to GEE and used to clip and filter all subsequent satellite processing.

Sentinel-2 Surface Reflectance Imagery was used for both the vegetation and wetness analyses, drawn from three periods: 1 July – 31 December 2018, 2019, and 2023. Using the same months in each year reduced the risk that the apparent land-surface change reflected normal seasonal vegetation or moisture cycles rather than a genuine change. Because the 2018 imagery contained substantial cloud contamination, the 2018 and 2019 observations were combined into a single, more robust baseline period. Clouds, cloud shadows, cirrus, and snow/ice were masked from all imagery before use.

3.2.1 Built-up Percentage (*BuiltPct*)

As previously mentioned, a suitable 2023 ISP percentage dataset was not available; instead, *BuiltPct*, representing the proportion of each analysis cell classified as built-up land rather than a direct physical measurement of impermeousness, was used. *BuiltPct* was derived from Google Dynamic World, a Sentinel-2-based 10 m land cover product available in GEE, using its built-up class. A July – December 2023 composite was created, and the modal land-cover label was used to classify each pixel as built (1) or not built (0). Dynamic World also provides a continuous built-probability layer, but this was used only for visual validation and confidence checking, since an average classification probability is not equivalent to the physical proportion of a cell covered by built land; only the binary classification was used to calculate *BuiltPct*. Both the binary and probability layers have been validated in QGIS for the value range, spatial plausibility, and correct NoData handling before use.

3.2.2 Vegetation Change (*NDVI Change*)

Vegetation greenness was measured using the NDVI calculated for Sentinel-2 as follows [37]:

$$NDVI = \frac{(B8 - B4)}{(B8 + B4)}, \quad (1)$$

where B8 and B4 represent near-infrared and red reflectance, respectively. NDVI was calculated per image prior to compositing (an NDVI-first approach), with NDVI changes calculated as $\Delta NDVI = NDVI_{2023} - NDVI_{baseline}$. A negative value was not automatically

interpreted as permanent deforestation, as the NDVI can vary for reasons other than permanent vegetation loss, including crop cycles, vegetation conditions, and moisture differences [38].

3.2.3 Wetness Decline (*WetDeclPct*)

Wetness was measured using the Modified Normalised Difference Water Index (MNDWI), calculated for Sentinel-2 as follows [39]:

$$MNDWI = \frac{(B3 - B11)}{(B3 + B11)}, \quad (2)$$

where B3 and B11 represent the green and short-wave infrared reflectance, respectively. The MNDWI was calculated per image and composited by median for the July – December periods of 2018 and 2019 (combined into a baseline) and 2023, using a common valid-data mask so that the change was calculated only where both baseline and 2023 observations were available. A pixel was classified as showing significant wetness decline where the baseline MNDWI exceeded -0.10 and the change in MNDWI was less than or equal to -0.10, producing a binary raster – a pixel-based grid, as opposed to the 200 m polygon grid used for modelling – of significant wetness decline (1) versus no significant decline (0).

3.2.4 Grid and Dataset Construction

The NDVI change, *BuiltPct*, and *WetDeclPct* rasters were aggregated to a 200 m grid using QGIS Zonal Statistics: the mean value per cell for NDVI change and the proportion of built/wetness-decline pixels ($\times 100$) for the two binary variables. The three new variables were added sequentially to the intermediate grids so that each addition could be validated individually before the final combined grid was produced. Grid-level values were transferred to the original 302 sample points using a spatial join, ensuring that the new predictors carried the same spatial meaning in both the training dataset (the 302 labelled points used to train the model) and the prediction dataset (the full 213,908-cell grid used to generate catchment-wide susceptibility estimates). The extraction of values directly from raster pixels at each sample point was deliberately avoided because it would introduce a scale mismatch between the training data and the 200 m prediction grid. Both datasets were checked prior to modelling to ensure that the original predictor variables remained intact, all three new variables were present and within the expected ranges and the training and prediction datasets shared the same predictor definitions.

3.3 Experimental Design and Results

Four enhanced-model experiments have been tested against the baseline as part of this study, with each experiment building incrementally on the previous one, so that changes in performance could be attributed to specific additions rather than simultaneous changes across multiple variables (Table 2). All models were evaluated using the same held-out test set, comprising 61 sample locations (30 non-flooded and 31 flooded). The evaluation metrics included flooded accuracy, precision, recall, F1-score, and Cohen's Kappa coefficient. Cohen's Kappa was included because it measures the agreement between predicted and observed flood classes while accounting for agreement that might occur by chance [40, 41]. Higher values of this coefficient indicate stronger agreement.

The baseline model achieved an overall accuracy of 92% and a Cohen's Kappa of 0.8363 for the 61-point test set using the original 10 flood-conditioning variables. This served as a reference point against which all enhanced configurations were compared.

Experiment 1 tested the direct substitution of BuiltPct for CN in the model. The resulting model reproduced the baseline results with 92% accuracy and 0.8363 Cohen's Kappa and identical per-class precision and recall. This suggested that for this test set, BuiltPct was able to replace CN without reducing the predictive performance, although identical results did not necessarily imply that the two variables contained the same underlying information.

Experiment 2 retained CN and added BuiltPct, producing the strongest results of all five configurations tested: 97% accuracy and a Cohen's Kappa of 0.9344. Relative to the baseline, the accuracy improved by 5 percentage points and Cohen's Kappa by ≈ 0.10 , representing the largest improvement observed across the experiments. Given that replacing CN with BuiltPct in Experiment 1 did not change model performance, the improvement observed when both variables were retained suggests that they may provide complementary information. CN appears to capture runoff-related soil and land cover characteristics that BuiltPct does not, while BuiltPct adds a more direct, current measure of built-up concentration that CN does not. Therefore, using both variables together appeared to improve discrimination beyond either variable used alone.

Experiment 3 added NDVI_Change to the model containing CN and BuiltPct, reaching 95% accuracy and Cohen's Kappa of 0.9017, 3 percentage points higher than the baseline but 2 percentage points lower than Experiment 2. Precision and recall both settled at 0.94 – 0.97 across classes, slightly less balanced than Experiment 2's near-uniform 0.97 scores. This indicates that adding vegetation

change information did not further improve the test set performance beyond the model containing CN and BuiltPct. This did not necessarily mean that NDVI change was unrelated to flood susceptibility; however, its information may have overlapped with other predictors already in the model, or its additional effect may have been too small to produce a measurable improvement within the relatively small 61-observation test set.

Experiment 4 added WetDeclPct to the variable set used in Experiment 3 and produced identical classification performance, with 95% accuracy, Cohen's Kappa of 0.9017, and matching class-level precision and recall. This performance remained higher than that of the baseline but slightly below that of Experiment 2. The addition of WetDeclPct did not improve classification on this test set. One possible explanation was the spatial sparsity of the wetness-decline signal, with only approximately 5000 – 6000 of the 213,908 prediction grid cells containing non-zero values. However, this sparsity may itself have reflected limitations in the threshold-based method used to derive the wetness-decline raster, rather than indicating that wetness decline is inherently unimportant for flood susceptibility. Therefore, the result should be interpreted cautiously, as it may reflect either limited additional predictive information or limitations in how wetness decline was represented. Future work could apply a sensitivity analysis using alternative baseline and MNDWI-decline thresholds to assess the robustness of the detected pattern and determine whether different threshold choices provide greater predictive value.

3.4 Visual Comparison of Baseline and Experiment 2

Moreover, the baseline and Experiment 2 model were compared using feature-importance plots and their respective generated flood susceptibility maps. Feature importance measures the contribution of each input variable to the machine-learning model's predictions, indicating which factors most strongly influence the resulting flood susceptibility map (Figure 1). The feature-importance results show that distance to the river (DTRiver) was the dominant predictor in both experiments. Its importance was approximately 0.36 in the baseline model and 0.34 in Experiment 2, indicating that proximity to the river remained the most influential factor after the additional predictor was introduced.

The relative contribution of the remaining variables changed between the two experiments. In the baseline model, DTRoad and Slope were the next most important variables, followed by TWI, DEM, NDVI, curvature and rainfall. Experiment 2 included the additional BuiltPct variable, representing the percentage of built-up land. Al-

Table 2: Machine Learning Experimental Configurations.

Configuration	Change from baseline	Purpose	Accuracy / Cohen's Kappa
Baseline	Original 10-variable model	Reproduced the original model and provided the reference result	0.92 / 0.8363
Experiment 1	CN replaced with BuiltPct	Tested whether built-up percentage could replace the existing curve-number representation	0.92 / 0.8363
Experiment 2	CN + BuiltPct	Tested whether built-up information added predictive information beyond CN	0.97 / 0.9344
Experiment 3	CN + BuiltPct + NDVI_Change	Tested whether vegetation change information improved performance beyond CN and BuiltPct	0.95 / 0.9017
Experiment 4	CN + BuiltPct + NDVI_Change + WetDeclPct	Tested the effect of integrating the full set of new human-impact and land-surface-change variables	0.95 / 0.9017

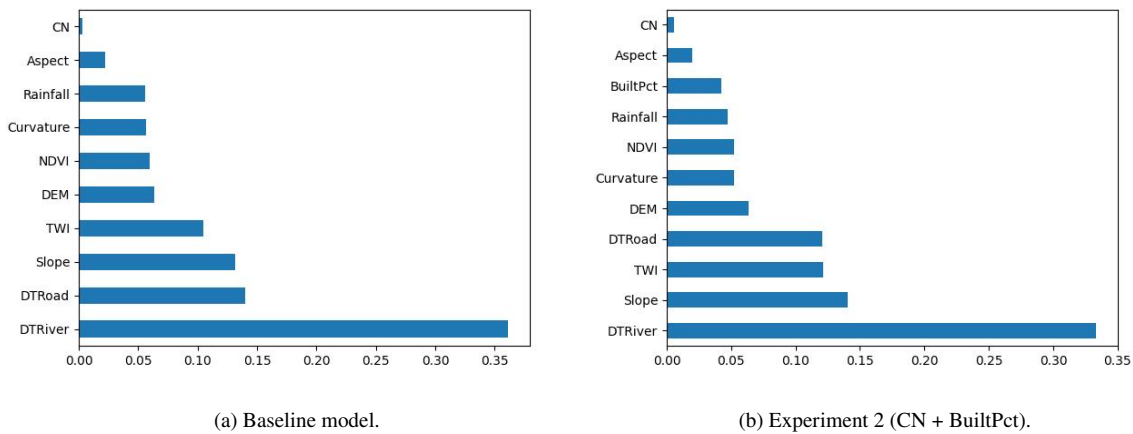


Fig. 1: Comparison of feature importance for the baseline model and Experiment 2.

though BuiltPct had a comparatively modest importance, its inclusion altered the ranking and contribution of the other variables. Slope, TWI, and DTRoad became more closely grouped in importance, while rainfall, NDVI, and curvature remained lower-ranking predictors. Aspect and CN had the smallest contributions in both experiments.

Furthermore, the corresponding flood susceptibility maps (not shown) appear visually similar, suggesting that adding BuiltPct did not substantially change the broad spatial pattern of susceptibility. However, visual similarity does not mean that the maps are identical. A pixel-level comparison of the Baseline flood susceptibility map and the Experiment 2 flood susceptibility map found that 58,568 of 276,477 pixels differed, representing 21.18% of the ren-

dered image. Of these, 49,709 pixels, or 17.98%, had an RGB difference magnitude of at least 5. The mean absolute RGB difference was 2.15, while the maximum difference in an individual colour channel was 226.

A flood susceptibility map difference overlay was created by using the baseline flood susceptibility map as the background and comparing it pixel by pixel with the Experiment 2 map (Figure 2). Pixels with a sufficiently large colour difference were highlighted by blending them with red, while pixels with little or no difference were left unchanged.

Red or reddish areas indicate locations where the Experiment 2 image differs noticeably from the baseline. The strength and shade of the red should be interpreted as an

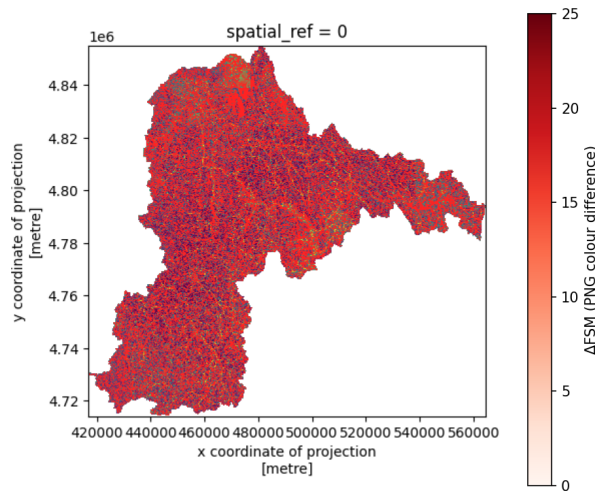


Fig. 2: Difference overlay between the Experiment 2 and baseline flood susceptibility maps. Red indicates areas of change; the background shows the baseline susceptibility map.

indication of where differences occur, not as a new flood-susceptibility value. Because this comparison uses rendered images, colour variations may also result from map rendering, labels, boundaries, or other visual elements.

These findings indicate that Experiment 2 produced measurable local changes while preserving the overall susceptibility pattern established by the baseline model. The stability of DTRiver as the dominant predictor and the strong visual similarity between the maps suggest that river proximity continued to control the broad distribution of flood susceptibility. The differences detected in the maps may reflect local changes associated with the inclusion of built-up land, but they should be interpreted cautiously because the comparison was performed on rendered PNG images. Consequently, some differences may result from map rendering, colour transitions, labels, boundaries, or anti-aliasing rather than changes in the underlying susceptibility values. This comparison therefore provides visual evidence of model-output variation; however, a direct comparison of the original numerical prediction rasters would be required to quantify changes in susceptibility more rigorously.

3.5 Future Improvements

Further improvement would be to evaluate the models using a larger independent test set. The current evaluation was based on only 61 observations; therefore, the performance metrics may be sensitive to the classification of a small number of points. Increasing the number and spa-

tial coverage of test observations would provide a more robust assessment of model generalisability. Nonetheless, given the consistency of the patterns observed across the four experimental configurations, the current results are considered a reasonable initial reflection of the added variables' impact, even though a larger test set would allow these findings to be confirmed with greater confidence.

While this model was developed and validated using data specific to Rwanda, the underlying variables (built-up percentage, vegetation change, and wetness decline) reflect patterns of human-driven land modification common to many regions globally facing similar flood and landslide risk [42,43]. Rwanda serves as a case study for a broader, extensible approach: with recalibrated thresholds and locally appropriate baseline data, the modelling framework presented here could be adapted to other regions facing comparable land-use-driven hazards.

4. Ground sensor development

4.1 Introduction

This section explores how electronic ground sensors can help to compensate for the low revisit cadence, limited resolution and long data processing timelines often experienced with satellite EO measurements as shown in Table 1 by providing additional data to feed into the software models. These sensors may also be used to detect tiny ground movements indicative of instability, which may not be picked up by spaced-based observation effectively. As part of this research, a prototype sensor has been developed to prove the concept and explore the challenges associated with deploying a network of sensors in remote areas.

4.2 Justification

This project depends on ground sensors installed in high risk and strategically important areas for two reasons. Firstly, when installed appropriately the ground sensors can measure environmental conditions with a high temporal and spatial resolution for locations where satellite data is either unavailable or unreliable. These measurements can also provide a source of ground truthing for data collected from EO satellites. However, high spatial resolution inherently requires many sensors spread across the target area recording data simultaneously for detailed analysis. The cost of manufacturing and installing each sensor is therefore critical, and is the main factor driving the cost of scaling the observation area. Secondly, the sensors themselves can monitor ground movement directly and send alerts if they detect instability, providing an additional layer of monitoring in high-risk areas. This is covered in

more detail in Section 5.

4.3 Existing Solutions

Numerous sensor systems have been proposed in literature that could be incorporated into this network, such as those listed extensively in the analysis by Bagwari *et al.* [44] or either of the designs presented by Chu *et al.* [45] and Ragnoli *et al.* [46]. However, even sensors that claim to be relatively cheap often cost several hundred USD each (for example, the Sitka project in [45] lists a cost of USD 940 per node). There are also industrial solutions on offer, from companies such as Geokon [47] and World Sensing [48], but the costs for these products appear to be based on individual quotations.

For targeted locations in affluent countries (or for industries like mining and building), these might be acceptable financial commitments. When working in the developing world and especially when planning systems to cover huge areas of land, these costs are not viable. For this project, price was consequently a driving factor for sensor design and compromise is necessary to keep the cost of each sensor as low as practical while maintaining functionality and reliability.

This project has also focused on using natural materials that are abundant in the local area for the sensor housings to reduce costs and minimise the environmental impact of the whole system. For example, bamboo culms could be processed into small cylindrical cases to contain the sensor components as a cheap and convenient housing if the natural degradation of the bamboo is tolerable. This approach supports the project's objectives of affordability, open-source development, and education by promoting local fabrication using readily available materials and basic tools, thereby reducing dependence on commercially manufactured components or expensive equipment such as 3D printers.

4.4 System Architecture

To achieve a low-cost design, the instrumentation was targeted to the most scientifically relevant measurements. Table 3 provides a list of the sensors chosen for this project, which are strong predictors of conditions related to a high risk of floods and landslides [7].

The table also specifies the auxiliary electronics for the sensor, such as the host microcontroller and radio module. For each of the devices, components were chosen which are available both as bare integrated circuits, and as ready-made breakout boards with all the necessary support circuitry already included. This provides several advantages.

First, it makes prototyping the sensor considerably easier because the breakout boards can be connected to the host microcontroller using jumper wires without soldering tiny components. Secondly, the sensor can be hand built using minimal equipment when used as an educational project in classrooms or community centres anywhere in the world.

Once the larger prototype sensor has been successfully demonstrated, the system can then be redesigned into a much more compact system using the individual components instead of the breakout boards. The breakout boards provide convenience for development but come with a cost premium, so switching to the components themselves reduces unit cost significantly. Keeping the same components across the two sensor designs means that the control software on the microcontroller can be consistent, helping to simplify the development process and improve code reusability.

Although this prototype sensor does not provide the data resolution or the number of readings that more complex sensors can (e.g. soil moisture at depth, water table, rainfall in [45]), the unit cost can be significantly lower. This allows many more sensor nodes to be manufactured within a limited budget, helping to increase the distribution and extend the reach of the project.

4.5 Sensor Design

4.5.1 Radio Communication

As minimising cost is one of the primary design objectives, it is important to maximise the area covered and operational lifetime of the sensor network while minimising the number of deployed nodes. Achieving this requires long-range communication with low power consumption. For this reason, the LoRa radio communication protocol was selected. To further improve network coverage and eliminate the need for every sensor node to have a direct internet connection, an open-source mesh network based on the LoRaMesher library was implemented [49,50]. Within this architecture, sensor nodes relay measurements through neighbouring nodes until they reach a gateway with internet connectivity, increasing the effective communication range, improving network resilience, and enabling flexible deployment in remote areas.

4.5.2 Power Considerations

As these sensors are planned for deployment in remote areas, each one needs to be fully self-sufficient. It would be too labour-intensive to manually replace batteries in lots of sensors, so a small solar panel was included in the design to recharge the battery. This adds an extra cost to

Table 3: Sensor measurements and component part numbers.

Device	Part number	Description
Sensors		
Temperature & humidity	SHT40	Ambient temperature and relative humidity
Soil moisture	Generic	Capacitive soil moisture probe
Accelerometer	LIS2DW12	Three-axis acceleration measurement
Support electronics		
Host microcontroller	ESP32-C3 Supermini	Low-power microcontroller with integrated Wi-Fi and Bluetooth
Real-Time Clock (RTC)	RV-3028	Ultra-low-power RTC with backup coin cell
LoRa radio module	DX-LR30	Long range data transmission
Battery	Generic	150 mAh, 3.7 V Lithium Polymer cell
Solar cell	Generic	80 mm x 45 mm, around 350 mW
Solar charger circuit	Generic	Interface between solar panel and battery

the sensor, but should allow much longer operation and a smaller battery, reducing the running costs of the overall system.

Radio transmission can be energy intensive, in comparison to the tiny amounts of power used to run the sensors themselves. For this reason, the radio module will be turned off for most of the time, and only activated when the microcontroller needs to send a message to the network.

4.5.3 Timekeeping

The Real-Time Clock (RTC) on the sensor keeps accurate time and allows the sensor to schedule various tasks. For example, the sensors can be coordinated to collect data at the same time each day and to turn on their radios for known periods to transmit the measurement data back to the base station.

4.5.4 Detecting Movement

The purpose of the accelerometer is to detect movement and allow the microcontroller to decide whether the motion is a short-term, unrelated event (for example an animal knocking into the sensor) or a significant incident like a landslide which requires alerting the relevant authorities. The thresholds for this decision will need to come from experimental data, with consideration to minimise false positives while ensuring that landslides are reported even if relatively small.

4.5.5 Design Documentation

The detailed design of the sensor is contained in the documentation available on the GitHub page for this project, linked from the website at kurinda.space. The sensor

and supporting system is still under development, so the live documentation will be updated regularly as the project progresses.

Figure 3a shows the Printed Circuit Board (PCB) design of the first version of the sensor, containing the components listed in Table 3. Next to it, Figure 3b shows the first assembled sensor.

4.5.6 Sensor cost analysis

The Bill Of Materials (BOM) cost including all the sensors listed in Table 3 was around GBP 50, plus GBP 40 per unit for the manufacture of the PCB. Manufacturing at larger scale should bring the unit cost down further (especially for the PCB itself), so it may be possible to produce a complete sensor for under USD 100 if cost-effective design decisions are taken. Once the more compact sensor is designed, this cost should come down significantly, as each sensor device costs on the order of a few USD instead of around USD 10 for the breakout boards.

4.5.7 Future Improvements

With the knowledge gained from the design and initial evaluation of the first sensor prototype, several opportunities for further development have been identified. A current limitation is the lack of a dedicated testing facility to quantify the performance and reliability of the individual sensing components under controlled conditions. Landslide simulators, such as that developed by Longoni *et al.* [51], provide an effective means of evaluating ground sensors and will be used in future work to validate the functionality of the proposed designs. Beyond validating the sensors themselves, future work will also focus on strengthening the integration of EO and ground-based sensing. In situ

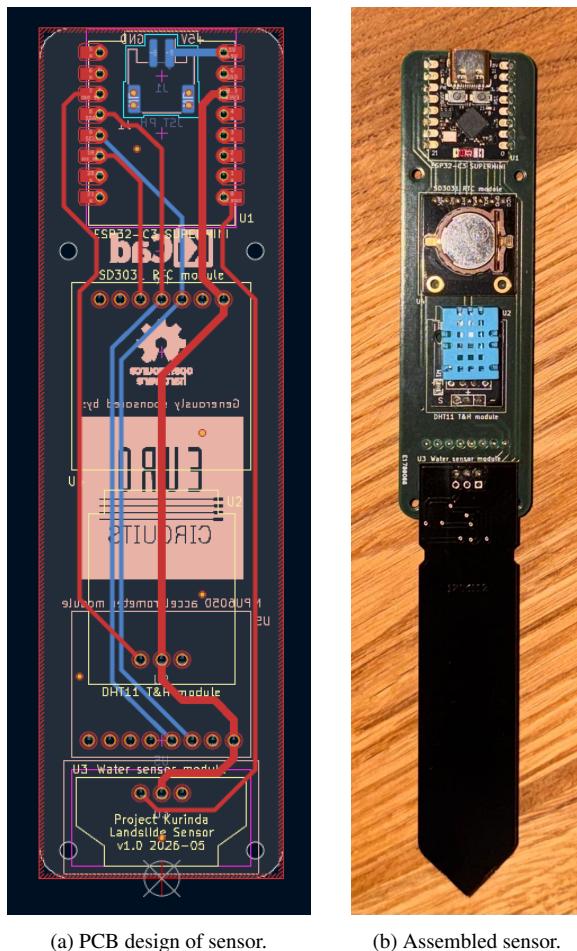


Fig. 3: Landslide sensor hardware.

measurements from the sensor network will be used to validate satellite-derived products and improve the interpretation of remotely sensed variables. For example, soil moisture sensors deployed in urbanising catchments can verify satellite-derived estimates of impervious surface expansion by comparison with measured infiltration behaviour, while sensors installed on deforested or mining-disturbed slopes can relate satellite-observed vegetation loss to measured ground movement. Incorporating these validated observations into the machine learning framework is expected to produce a more accurate susceptibility model that better captures the influence of human-driven land modification than existing approaches in Rwanda. This directly addresses a documented limitation of the country's current landslide monitoring, which relies largely on manual coordination with limited integration of remote sensing, automation, and predictive modelling [52]. The result-

ing susceptibility outputs are intended to feed directly into the proposed EWS and community-response framework described in Section 5.

5. Early Warning System (EWS)

The research presented in this study, together with the improvements identified through the integration of ground-based sensing and EO, demonstrates an approach that can be applied to regions worldwide that are exposed to flooding and landslide hazards. However, producing an accurate prediction through the developed ML model is only one part of an effective EWS. The information generated by the system must also reach the communities at risk in a form that is accessible, understandable and actionable.

5.1 Integration into Society

The output from the ML model discussed in Section 3 provides an estimation of the likelihood of a flood or landslide occurring, while ground sensors act as an immediate fail-safe in case of an unpredicted catastrophe. A clear link is therefore required between the model output/ground sensor readings and the communities that need to act on this information. To address this, an online platform has been developed to communicate information about the warning system and additional detail on the associated ground sensors, such as ID, location, status, and current readings. The platform is intended to support the monitoring of potential floods and landslides in remote areas while providing communities with clear information on the risks identified by the system. Importantly, this set-up only requires access to a single mobile device within a community, which would then allow the information to be shared more widely. This is particularly important in remote regions where mobile device access remains limited.

The output from the hybrid monitoring network is categorised using a traffic light system to provide a simple representation of the predicted hazard risk, as outlined in Table 4.

This approach provides a simple and widely recognisable method of communicating the hazard monitoring output without requiring communities to interpret the complex scientific information behind the prediction. This is particularly important in remote communities where technical information may not always be trusted or easily understood [53]. By presenting the output as a clear risk level with a defined response, the system aims to make the information more accessible and increase the likelihood that a warning results in appropriate action.

Where a high or extreme risk is identified, the system

Table 4: Traffic light system to categorise hazard likelihood based on ML model outputs and recommended community responses.

Hazard likelihood	Recommended response
Low	No immediate action required.
Medium	Take initial precautionary measures and begin preparing.
High	Take immediate action to prepare and inform the wider community.
Extreme	Evacuate to a safe location and place emergency services on standby.

would trigger an SMS alert to mobile phones within the identified at-risk area. SMS is important for distributing urgent warnings because internet and mobile data networks may be unavailable in remote regions or may become unreliable during a hazard event. Using simple existing cellular network infrastructure would therefore provide a direct method of communicating warnings to affected communities.

Furthermore, it is important to provide communities with the knowledge and education required to understand what actions should be taken before a hazard occurs, how to respond when an alert is received and what actions should be taken following an event. This links hazard prediction with preparing and response rather than treating early warning as solely a prediction problem. As the frequency and severity of some flood and landslide hazards are expected to increase as a result of climate change, the development of accessible and scientifically informed EWSs is increasingly important for vulnerable communities worldwide. Section 5.2 demonstrates the relevance of this integrated approach through a case-study into the national response and actions taken in Rwanda.

5.2 Proof of Application : Rwanda

A 2015 National Risk Atlas Report for Rwanda found that it is strongly susceptible to natural disasters, with 40% of the country's population having a moderate to very high level of susceptibility to landslides, as shown in Figure 4, and 43% of health facilities in the country facing a high level of susceptibility [54]. The country faces a growing landslide crisis, particularly in its western and northern highlands, where clay-rich soils, steep slopes, and intense rainfall create unsafe conditions [52].

With an escalating threat, Rwanda's disaster risk management relies upon fragmented and largely manual monitoring methods, with limited integration of remote sensing, automation, or predictive modelling [52]. Current land-

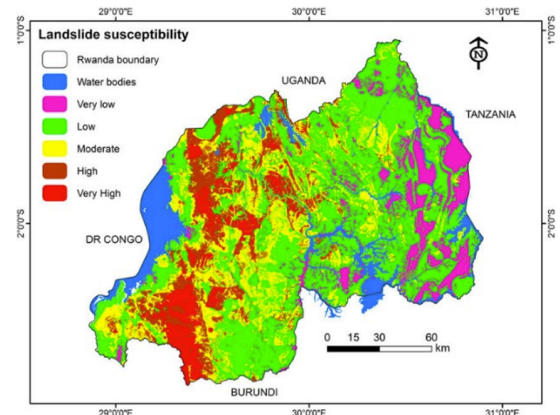


Fig. 4: Landslide susceptibility map of Rwanda produced by Spatial Multi-Criteria Evaluation [52].

slide monitoring systems in Rwanda are primarily reactive and lack real-time forecasting and predictive accuracy.

The consequences of this technological and procedural gap are significant. Without accurate early alerts, communities remain highly vulnerable [52]. The resulting losses include increased poverty in the affected regions, frequent displacement, disrupted public services, and a large financial strain on the national disaster recovery system [55].

An EWS is a crucial tool for building resilience and for safeguarding lives and livelihoods in a world of increased climate change, impacting communities around the world [56]. An EWS is only effective if it actively places people at its centre and ensures that all elements of the system prioritise those at risk [57].

Rwanda does not have a specific response plan for landslides, but instead places landslide threat under a single early warning framework in the national guidelines for emergency response and recovery. The guidelines categorise all hazards into four levels [58]. According to these guidelines, warning is defined as the transmission of accurate and timely information before harm occurs. Meteo Rwanda, the national meteorological agency, and the Rwanda Water Board are tasked with the detection of a hazard, and local disaster communities assess the damage and hand that information over to local community leaders [58].

Warning follows a consistent three-stage process: hazard monitoring, local verification, and community dissemination. Institutional ownership escalating from sector and district committees at hazard Levels I and II to MINEMA and national ministries at Level III, to national government supported by international bodies at Level IV [58]. This structure provides clear, severity-based accountability for

issuing and acting on landslide warnings, although stops short of specifying a dedicated technical monitoring mechanism for landslide precursors comparable to the hydrological monitoring in place in Rwanda. These approaches are reactive, can be imprecise, and they also cannot process large-scale, real-time data. This can lead to delayed warnings and inadequate risk communication [59].

In the 2024 “*National Action Plan for Building Country Resilience to Geohazards*”, Rwanda states that it has no baseline for landslide community awareness and rated the priority high [60].

In a proactive effort to reduce the impact of natural disasters, Rwanda has strengthened the capacity of frontline responders and invested in its financial resilience (RBC, 2025) [61]. While these initiatives mark meaningful institutional progress, they remain centred on training, coordination, and financial preparedness rather than on the technical monitoring infrastructure needed to detect landslide precursors directly. Rwanda must ensure that measures are in place to communicate risks and enhance public disaster response and preparedness for the future. Equal importance should be given to the social components (e.g. community involvement, communication) as to the technical aspects of an EWS [62].

6. Summary and Conclusions

To address the current lack of human-driven land change in hazard prediction models, a hybrid disaster-risk evaluation system consisting of publicly available, ML processed EO satellite data and self-designed low-cost, open-source ground sensors is proposed to combat flood and landslide hazards. The current EO satellite state-of-the-art was reviewed, and additional factors contributing to increased flood and landslide risk were discussed. The ground sensors were designed with affordability, simplicity, ecological suitability, and the validation of remote sensing data as the primary design objectives. Two variants are presented: a compact, mass-manufacturable PCB design and a second design intended for local fabrication using readily available tools and materials, offering greater educational value and autonomy. The ML model was based on and compared to a previously reported model applied to the Nyabarongo catchment in northwest Rwanda and investigated the addition of three variables directly related to human land modification. The experiment output from the model developed in this study showed largest improvement from the addition of the Built Percentage increased the baseline models accuracy from 92 to 97% and raised Cohen’s Kappa by approximately 0.10. Two other variable implementations (NDVI Change, Wetness Decline) slightly raised the accu-

racy compared to the baseline, but not to the same extent as the Built Percentage. These results demonstrate that human activity has a significant influence on predictive accuracy and, consequently, on flood and landslide risk, highlighting the importance of incorporating anthropogenic factors into natural hazard risk assessments. Future versions of the adjusted predictive model should focus on larger testing datasets and replacing binary variables with more descriptive or continuous representations where appropriate. A proposed traffic light warning system offers a link between the monitoring system output and the communities that need to act on this information. SMS alerts are able to reach vulnerable groups to provide warning to at risk groups. A review of current EWS implementation in Rwanda is presented to highlight the relevance of this work. This study contributes to the development of low-cost, open-source technologies that support climate change adaptation and disaster risk reduction in vulnerable regions worldwide.

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